

Thyagaraja M

Frontend | Backend Developer| Full Stack| Ai

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Professional Summary

Full-Stack Developer with 2+ years of hands-on experience designing and deploying scalable web and mobile applications. Expertise in Frontend and backend development , HTML, CSS, JavaScript, React, Tailwind, TypeScript, and Node js, Express js, AI with proven skills in database management (SQL - mysql, PostgreSQL) and cloud deployment (GCP).

Education

Bachelor of Computer Applications (BCA) - IGNOU (Distance Learning), Pursuing 3rd Semester
ITI Mechanical - Industrial Training Institute Bangalore (2001-2003)
Full Stack Web Development Course (Completed 2024)

Technical Projects

1. AI Home Decorator (Full-Stack Web & Mobile)

- Designed and developed a full-stack application using React, TypeScript, and Node.js. Implemented Firebase services and Supabase for auth/database. Deployed on Google Cloud.

- [Web Application](#) | [Mobile Application](#)

2. Wall Paint AI (Full-Stack Web)

- Built an AI-driven wall paint visualization platform using React and TypeScript. Managed real-time data storage and auth via Firebase.

- [Web Application](#)

3. Dream Home Calculator (PWA & TWA)

- Developed a responsive Progressive Web App using HTML, CSS, and JS. Implemented Trusted Web Activity for Google Play Store deployment.

- [PWA Web Application](#) | [Mobile TWA](#)

4. HDE

- Developed full-stack functionality with React, TypeScript, and Node.js integrated with Supabase.

- [Web Application](#)

Skills & Expertise

Programming Languages & Web Technologies: C, C++, JavaScript, React, TypeScript, HTML, CSS, Node.js.

Backend: MySQL, Supabase, Google Cloud Platform (GCP).

Product & Marketing: App Deployment (Play Store), SEO (Search Console, XML Sitemaps), Google Ads, Digital Marketing.

Design & Visualization: AutoCAD, 3D Modeling (Interior/Exterior), Realistic Rendering, Product Design. [[3D Portfolio](#)]

Experience

Eteam (Client: Lowe's India) | 3D Artist | Sep 2021 - Nov 2022

Created product-based images for e-commerce, including interior, paint, and flooring automation.

Dubai Expo 2020 | 3D Designer | Aug 2019 - May 2020

Concept design, 3D modeling of pavilions, and CAD drafting for thematic elements.

IMG Worlds of Adventure - Dubai | 3D Designer | Nov 2013 - Jun 2017

Thematic and lighting design development for large-scale entertainment projects.

Technical Projects

1. [AI Home Decorator](#) (Full-Stack Web & Mobile)

Overview: An end-to-end spatial design platform allowing users to visualize interior transformations in real-time.

- **AI Integration:** Leveraged Generative AI models and Google AI Studio to implement a "Snap, Select, Decorate" workflow. I engineered prompts and API pipelines to handle image-to-image transformations, ensuring stylistic consistency in generated room layouts.
- **Frontend & Backend:** Developed a high-performance frontend using React and TypeScript with Tailwind CSS. Built a scalable backend using Node.js and Supabase for real-time data synchronization and user authentication.
- **Complexity & Solution:** The primary challenge was managing large image assets and low-latency AI responses. I solved this by implementing an optimized client-side state management system and deploying the application on Google Cloud Platform (GCP) to ensure reliable uptime and fast processing.

2. [HDE](#) - Construction Estimator & Directory (SaaS)

Overview: A comprehensive platform for construction cost estimation and a verified professional directory.

- **Core Functionality:** Built a logic-heavy estimation engine that calculates project costs based on real-time material data. Integrated Supabase for complex relational database management.
- **Frontend & Deployment:** Developed a responsive Progressive Web App (PWA) using React.
- **Complexity & Solution:** Faced significant routing challenges when deploying to GitHub Pages (404 errors on refresh). I engineered a custom SPA routing fallback and transitioned the project from HashRouter to standard routing with a 404.html redirection logic to maintain a seamless UX. I also implemented a "Verified Professional" badge system using RBAC (Role-Based Access Control).

3. [Wall Paint AI](#) (Visualization Platform)

Overview: An AI-driven tool focused on real-time color visualization for architectural surfaces.

- **AI & Visualization:** Used my 10-year background in 3D visualization to guide the AI's understanding of lighting and shadows. The tool allows users to overlay different paint textures and colors onto uploaded photos dynamically.
- **Tech Stack:** Built with React, TypeScript, and Firebase.
- **Complexity & Solution:** The difficulty in achieving realistic color blending on irregular wall surfaces. I solved this by combining CSS masking techniques with AI-driven edge detection, ensuring the paint colors respect the natural shadows and lighting of the room.

I have architected and integrated a **Professional Directory** (homedesignenglish.com/directory) using a robust **Supabase/Node.js** backend. My strategic roadmap focuses on scaling this module into [India's premier directory](#) for construction and design professionals, leveraging automated verification workflows and AI-driven search to ensure high-quality listings.